

CHAPTER 2

BASIC ALGEBRAIC PROPERTIES OF COMPLEX NUMBERS

1. Which of the following correctly states the commutative property under addition for complex numbers?

A) $z_1 + z_2 = z_1 - z_2$

B) $z_1 \cdot z_2 = z_2 + z_1$

C) $z_1 + z_2 = z_2 + z_1$

D) $z_1 + z_2 = z_1 \cdot z_2$

2. Which expression correctly shows the associative property of multiplication for complex numbers?

A) $(z_1 \cdot z_2) + z_3 = z_1 \cdot (z_2 + z_3)$

B) $(z_1 \cdot z_2) \cdot z_3 = z_1 \cdot (z_2 \cdot z_3)$

C) $z_1 \cdot z_2 = z_2 \cdot z_1$

D) $z_1 \cdot (z_2 + z_3) = z_1 z_2 + z_1 z_3$

3. What is the additive identity for complex numbers?

A) 1

B) i

C) $0 + 0i$

D) $1 + 0i$

4. What is the multiplicative identity for complex numbers?

A) 0

B) i

C) $0 + i$

D) $1 = 1 + 0i$

5. For which values of z does the multiplicative inverse z^{-1} NOT exist?

- A) When z is purely real
- B) When $z = 0$
- C) When z is purely imaginary
- D) When z has a negative real part

6. If $z = 3 + 4i$, what is the denominator used in computing z^{-1} ?

- A) 7
- B) 12
- C) 25
- D) 1

7. Which property directly tells us that $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$?

- A) Commutative property of addition
- B) Distributive property
- C) Associative property of addition
- D) Additive identity

8. Division of z_1 by z_2 ($z_2 \neq 0$) is formally written as:

- A) $z_1 - z_2$
- B) $z_1 \cdot z_2$
- C) $z_1 \cdot z_2^{-1}$
- D) $z_1 + (-z_2)$

9. What does the commutative property under multiplication state?

- A) $z_1 + z_2 = z_2 + z_1$
- B) $z_1 \cdot z_2 = z_1 + z_2$
- C) $z_1 \cdot z_2 = z_2 \cdot z_1$
- D) $z_1 \cdot (z_2 \cdot z_3) = (z_1 \cdot z_2) \cdot z_3$

10.If $z = x + iy$, which expression gives z^{-1} ?

A) $x/(x^2+y^2) + i \cdot y/(x^2+y^2)$

B) $x/(x^2+y^2) - i \cdot y/(x^2+y^2)$

C) $-x/(x^2+y^2) + i \cdot y/(x^2+y^2)$

D) $1/(x+y)$

ANSWERS

QUESTION NO	ANSWER OPTION
1)	C)
2)	B)
3)	C)
4)	D)
5)	B)
6)	C)
7)	C)
8)	C)
9)	C)
10)	B)